PART C – NETWORK ANALYSIS

Scenario 1:

Laptop IP: 192.168.1.25

Answer: It is a Private IP Address.

Explanation: The IP address 192.168.1.25 belongs to the private IP address range (192.168.0.0 – 192.168.255.255). It is used for communication within a local network (LAN). It cannot be accessed directly from the Internet.

Scenario 2:

Mobile receives 192.168.1.100 after connecting to WiFi. Which service assigned it?

Answer: DHCP (Dynamic Host Configuration Protocol).

Explanation: DHCP automatically assigns IP addresses to devices when they connect to a network. The WiFi router usually acts as a DHCP server and provides the mobile with an IP address, subnet mask, default gateway, and DNS details.

Scenario 3:

Server IP: 49.205.100.25. Can Internet users directly access it? Why?

Answer: Yes, Internet users can potentially access it directly.

Explanation: 49.205.100.25 is a Public IP address. Public IP addresses are globally routable on the Internet. However, access depends on firewall rules, security settings, and whether the required services and ports are open.

Scenario 4:

Laptop cannot access websites outside the local network. Which network component should be checked first?

Answer: Router (Default Gateway).

Explanation: The router connects the local network (LAN) to the Internet (WAN). If the default gateway or router is not working properly, the laptop may communicate with local devices but cannot access external websites.

Scenario 5:

A DevOps Engineer deploys EC2 Instance, Docker Container and Kubernetes Pod. Why does each require an IP Address?

Answer: Each requires an IP address for network communication and connectivity.

Explanation:

1. EC2 Instance: Requires an IP address to communicate with other servers and access the Internet.
2. Docker Container: Requires an IP address to communicate with other containers and external networks.
3. Kubernetes Pod: Requires an IP address to communicate with other Pods and Kubernetes services.

Conclusion: IP addresses help identify devices and resources on a network and allow them to send and receive data.